35. On Maximal Domains of Integral Functions

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The following investigations are closely related to Hilbert's problem on relatively integral functions. They arose from an occasional question of E. Hecke, who had been able to settle a special case directly by computation because he needed it as a lemma.

The question is this. By a maximal domain of integral functions in $x_1, \ldots, x_r$, namely polynomials, power series, or quotients of such, inside a given domain $\mathfrak{B}$, I mean one which can no longer be enlarged by adjoining integral denominators: whenever $ag(x)$ belongs to it, where $g(x) \in \mathfrak{B}$ and a is a nonzero rational integer, then $g(x)$ itself also belongs to it. Every integral domain $\mathfrak{A}$ in $\mathfrak{B}$ determines uniquely a maximal domain $M(\mathfrak{A})$; it consists of all functions $g(x) \in \mathfrak{B}$ for which there is at least one $a \neq 0$ such that $ag(x) \in \mathfrak{A}$.

The question is what can be said about the denominators a of this maximal domain $M(\mathfrak{A})$ when the original integral domain $\mathfrak{A}$ is finite, that is, when it consists of all integral polynomials in finitely many functions $f_1(x), \ldots, f_s(x)$ from $\mathfrak{B}$. Here $\mathfrak{B}$ is to be taken as all integral polynomials or power series in $x_1, \ldots, x_r$, or as quotients whose only denominators are those occurring in the $f_i(x)$ and their powers.

I show that, in the case of power series or their quotients subject to the additional hypothesis that only algebraic dependencies occur among the $f_i(x)$, the a may always be chosen so that only finitely many different prime numbers divide them, although in general to arbitrary powers. The latter point is immediately clear: if $ag(x)$ belongs to $\mathfrak{A}$, but $a^q g(x)^q$ does not for any exponent q, then all powers a^q occur, for example in $\mathfrak{A} = [2x]$.

The question of boundedness can only be formulated as follows: can one give a generally infinite system of generators for $M(\mathfrak{A})$ such that, in the associated denominators, the exponents of the prime numbers remain bounded? Since only finitely many primes occur, this question is, by known arguments, identical with the question of the finiteness of $M(\mathfrak{A})$, hence with Hilbert's problem on relatively integral functions in its integral form. I cannot decide that question by the methods used here.

The proof that only finitely many different primes p occur in the denominators a rests on the fact that to every such p there corresponds at least one relation modulo p among the functions $f_i(x)$ generating $\mathfrak{A}$, a relation with integral coefficients which is not identically satisfied in x. Equivalently, let $\mathfrak{o}$ denote the integral polynomial domain in indeterminates $u_1, \ldots, u_s$. Then $\mathfrak{A}$ is a homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$. This homomorphism is mediated by a prime ideal $\mathfrak{a}$ of $\mathfrak{o}$, so that $\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}$. Similarly $\mathfrak{B}_p$ is a homomorphic image of $\mathfrak{o}_p$, the subscript denoting reduction modulo p. This homomorphism is mediated by a prime ideal $\mathfrak{c}_p$ of $\mathfrak{o}_p$, containing $\mathfrak{a}_p$. The prime p occurs in a denominator of $M(\mathfrak{A})$ if and only if $\mathfrak{c}_p$ is a proper overideal of $\mathfrak{a}_p$. This can happen only if either the transcendence degree of $\mathfrak{A}_p$ drops relative to that of $\mathfrak{A}$, or $\mathfrak{a}_p$ loses its property of being prime. Only finitely many primes can do this, since even modulo p an algebraic dependence implies the vanishing of the functional determinant, and since $\mathfrak{a}$ is an absolute prime ideal and so loses primality modulo only finitely many p.

All the arguments remain valid, with slight modifications, if rational integers are replaced by the integers of a finite algebraic number field and prime numbers by its prime ideals.

§1. Functional Determinants over Coefficient Domains of Characteristic p

Let $f_1(x_1, \ldots, x_r), \ldots, f_t(x_1, \ldots, x_r)$ be rational functions, or more generally formal power series or quotients of such, with coefficients in a field P of characteristic p. The derivatives $\partial f_i / \partial x_j$ are also to be formally defined, with the usual rules of calculation preserved. Let Ω be an algebraically closed extension field of P. Then, just as in characteristic zero:

If there is an algebraic dependence among the $f_i(x)$ with coefficients in Ω, then the rank of the functional matrix

$$\left(\frac{\partial f_i}{\partial x_j} \right) \qquad (1 \leq i \leq t,\ 1 \leq j \leq r)$$

is less than t.

In the polynomial domain $\Omega[u_1, \ldots, u_t]$, let $\mathfrak{m}$ denote the prime ideal of all polynomials $G(u)$ for which $G(f) = 0$ identically in x. By hypothesis $\mathfrak{m} \neq 0$. Let $G(u) \neq 0$ be a polynomial of least degree in $\mathfrak{m}$; in particular G is irreducible. As usual one obtains

$$\sum_{i=1}^{t} \frac{\partial G}{\partial u_i}(f) \frac{\partial f_i}{\partial x_j} = 0, \qquad j = 1, \ldots, r.$$

Hence either the rank of the functional matrix is less than t, or all $(\partial G/\partial u_i)(f)$ vanish. Since G was chosen of least degree, it follows that all $\partial G/\partial u_i$ vanish. Thus

$$G(u) = H(u_1^p, \ldots, u_t^p).$$

Because Ω is algebraically closed, hence perfect, one further obtains $G(u) = (T(u))^p$, contradicting the choice of G. The proof also applies in characteristic zero, with the last step omitted.

Corollary. Let $F_1(x), \ldots, F_t(x)$ be integral functions, i.e. polynomials, power series, or quotients of such, whose functional matrix has rank exactly t. Choose the prime p so that it divides neither the denominators of the $F_i(x)$ nor all t-rowed determinants of the functional matrix. Then $F_1(x), \ldots, F_t(x)$ remain algebraically independent modulo p. Indeed, the functional matrix of the reduced functions is obtained by reducing the coefficients modulo p, and by the choice of p it still has rank t.

§2. Formulation of the Theorem

As in the introduction, let $\mathfrak{A}$ be an integral domain of integral functions in $x_1, \ldots, x_r$, formed from polynomials or power series with rational integral coefficients, or quotients of such. In the case of power series, they may be formal or convergent in some region; only the fact that the functions form an integral domain is used.

Let Q be the quotient field of $\mathfrak{A}$, and let $\mathfrak{B}$ be an integral domain between $\mathfrak{A}$ and Q. As above, $\mathfrak{M}$ is called maximal in $\mathfrak{B}$ if from $ag(x) \in \mathfrak{M}$, with $a \neq 0$ integral and $g(x) \in \mathfrak{B}$, it always follows that $g(x) \in \mathfrak{M}$. The maximal domain $M(\mathfrak{A})$ generated by $\mathfrak{A}$ in $\mathfrak{B}$ is the intersection of all maximal domains in $\mathfrak{B}$ containing $\mathfrak{A}$, and it consists of all $g(x) \in \mathfrak{B}$ for which some $a \neq 0$ satisfies $ag(x) \in \mathfrak{A}$.

Assume now that $\mathfrak{A}$ is a finite integral domain with identity, with $f_1(x), \ldots, f_s(x)$ as an integral basis; thus $\mathfrak{A}$ consists of all integral polynomials in the $f_i(x)$. Let $\mathfrak{B}$ consist of all integral polynomials or power series in x, or of quotients whose only denominators are those occurring among the finitely many $f_i(x)$. Assume also that the variables x actually occur in at least one of the $f_i(x)$; otherwise there is nothing to prove.

For polynomials or rational functions the quotient field Q has transcendence degree t, $1 \leq t \leq r$, over the field K of rational numbers. If, for instance, $f_1(x), \ldots, f_t(x)$ form an irreducible system, then Q is a finite algebraic extension of $K(f_1(x), \ldots, f_t(x))$. Also, the functional matrices of $f_1, \ldots, f_t$ and $f_1, \ldots, f_s$ both have rank t.

For power series or their quotients we impose the additional hypothesis that this structure of Q remains valid: if t is the rank of the functional matrix of $f_1, \ldots, f_s$ and also that of $f_1, \ldots, f_t$, then Q is again to be a finite algebraic extension of $K(f_1(x), \ldots, f_t(x))$.

Now let $\mathfrak{o}$ be the integral polynomial domain in $u_1, \ldots, u_s$. Then $\mathfrak{A}$ is a ring-homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$. Therefore $\mathfrak{A}$ is ring-isomorphic to $\mathfrak{o}/\mathfrak{a}$, where $\mathfrak{a}$ is a prime ideal. The theorem is:

Theorem. Under the added hypothesis just stated, let $\mathfrak{o}_p, \mathfrak{a}_p, \mathfrak{A}_p, \mathfrak{B}_p$ denote the domains obtained from $\mathfrak{o}, \mathfrak{a}, \mathfrak{A}, \mathfrak{B}$ by passing to residue classes modulo p. Then, except for at most finitely many prime numbers p, the homomorphism from $\mathfrak{o}_p$ to $\mathfrak{A}_p$ is mediated by $\mathfrak{a}_p$; hence

$$\mathfrak{A}_p \simeq \mathfrak{o}_p/\mathfrak{a}_p \simeq \mathfrak{o}/(\mathfrak{a}, p\mathfrak{o}).$$

Corollary. If p is chosen so that $\mathfrak{A}_p \simeq \mathfrak{o}_p/\mathfrak{a}_p$, then $pg(x) \in \mathfrak{A}$ and $g(x) \in \mathfrak{B}$ imply $g(x) \in \mathfrak{A}$. Equivalently, $\mathfrak{A}$ is maximal in $\mathfrak{B}$ with respect to all primes except the finite exceptional set. Iterating gives: if a is divisible by none of the exceptional primes, then $ag(x) \in \mathfrak{A}$ and $g(x) \in \mathfrak{B}$ imply $g(x) \in \mathfrak{A}$.

§3. Proof of the Theorem

Lemma. The prime ideal $\mathfrak{a}$ mediating the homomorphism between $\mathfrak{o}$ and $\mathfrak{A}$ is an absolute prime ideal.

Let $\mathfrak{o}^*$ be the polynomial domain in $u_1, \ldots, u_s$ with arbitrary algebraic numbers as coefficients, and let $\mathfrak{a}^*$ be the ideal in $\mathfrak{o}^*$ generated by $\mathfrak{a}$. We must prove that $\mathfrak{a}^*$ is prime. It is enough to show that $\mathfrak{a}^*$ mediates the homomorphism from $\mathfrak{o}^*$ into the integral domain $\mathfrak{A}^*$ consisting of all polynomials in the $f_i(x)$ with algebraic coefficients.

Let $H(u) \in \mathfrak{o}^*$ and $H(f_1(x), \ldots, f_s(x)) = 0$. Let $\omega_1, \ldots, \omega_m$ be a linearly independent basis of the finite number field generated by the coefficients of H. After multiplying by a nonzero rational integer a, we may write

$$aH(u) = \omega_1 H_1(u) + \cdots + \omega_m H_m(u),$$

with $H_i(u) \in \mathfrak{o}$. From $H(f)=0$ follows

$$\omega_1 H_1(f) + \cdots + \omega_m H_m(f) = 0,$$

and hence $H_i(f)=0$ for all i. Thus all $H_i(u)$ lie in $\mathfrak{a}$, and $H(u) \in \mathfrak{a}^*$.

The proof of the theorem can now be phrased as follows. Choose p so that

$$\text{I. } \mathfrak{A}_p \text{ is defined,} \qquad \text{II. } \mathfrak{a}_p \text{ remains prime,} \qquad \text{III. } \mathrm{Trdeg}(\mathfrak{A}_p) = \mathrm{Trdeg}(\mathfrak{A}).$$

Then the homomorphism from $\mathfrak{o}_p$ to $\mathfrak{A}_p$ is mediated by $\mathfrak{a}_p$. Apart from finitely many exceptions these conditions hold. I is clear; II follows from the lemma and the fact that an absolute prime ideal loses primality modulo only finitely many primes; III follows from the corollary on functional determinants.

Let the homomorphism from $\mathfrak{o}_p$ to $\mathfrak{A}_p$ be mediated by a prime ideal $\mathfrak{c}_p$. Since $\mathfrak{a}_p \subseteq \mathfrak{c}_p$, the two ideals are equal precisely when their transcendence degrees agree. By the added hypothesis $\mathfrak{a}$ has transcendence degree t. Number the u_i so that the classes of $u_1, \ldots, u_t$ modulo $\mathfrak{a}$ form an irreducible system and the classes of $u_{t+1}, \ldots, u_s$ are algebraic over them. Then $\mathfrak{a}$ contains primitive integral polynomials

$$G_{t+1}(u_1, \ldots, u_t, u_{t+1}), \ldots, G_s(u_1, \ldots, u_t, u_s),$$

which reduce modulo p to nonzero polynomials. In each such polynomial the last variable actually occurs. Otherwise there would be an algebraic dependence among the classes of $u_1, \ldots, u_t$ modulo $\mathfrak{a}_p$. Therefore $\mathfrak{a}_p$ has transcendence degree t, is identical with $\mathfrak{c}_p$, and the theorem follows.

§4. Extension to Integral Algebraic Coefficients

The arguments remain valid, with slight changes, if rational integers are replaced by the integers of a finite algebraic number field K, and prime numbers by its prime ideals.

In the proof one must add to the exceptional prime ideals those which occur in the coefficients of the polynomials $G_{t+1}, \ldots, G_s$, since these can no longer be assumed primitive. This ensures that their reductions do not vanish. The condition of absolute irreducibility is unchanged.

The final corollary is then: if the integer a of K is divisible by none of the finitely many exceptional prime ideals, then $ag(x) \in \mathfrak{A}$ and $g(x) \in \mathfrak{B}$ imply $g(x) \in \mathfrak{A}$. If $\lambda_1, \ldots, \lambda_m$ are integers of K each divisible by exactly one exceptional prime ideal, but not by its square and not by the others, then it is enough to use denominators which are products of powers of these λ_i.

Let

$$(a) = \mathfrak{p}_1^{\alpha_1} \cdots \mathfrak{p}_m^{\alpha_m}$$

be the prime ideal decomposition of a, with the $\mathfrak{p}_i$ non-exceptional. Passing to the quotient ring R defined by (a), the ideals $\mathfrak{p}_i$ remain prime but become principal, and the factorization of (a) is unchanged. The proof of the corollary therefore remains as before: if $ag(x) \in \mathfrak{A}$, then after clearing a denominator prime to a, one obtains $g(x) \in \mathfrak{A}$.

For the statement about denominators, let R^* be the quotient ring in K defined by the finitely many exceptional prime ideals. In R^* every a is, up to a unit, a product of powers of the basis elements λ_i. Returning to integers gives

$$\gamma a = \beta\, \lambda_1^{\alpha_1} \cdots \lambda_m^{\alpha_m},$$

where γ, β are divisible by none of the exceptional prime ideals. Hence from $ag(x) \in \mathfrak{A}$ it follows that

$$\lambda_1^{\alpha_1} \cdots \lambda_m^{\alpha_m} g(x) \in \mathfrak{A}.$$

Notes. The finiteness results used here are the standard Hilbert–Noether finiteness theorems. For integral projective invariants of a system of ground forms, the result says that they can be represented by a full invariant system in the usual sense, with only finitely many different primes in the denominators. The transcendence degree of a system is the invariant number of algebraically independent functions on which the system algebraically depends; the transcendence degree of a prime ideal is that of its residue field. The absolute primality assertion is reduced to the known theorems on absolute irreducibility and elimination theory. For functional determinants in characteristic zero one may also use van der Waerden's purely algebraic proof. The additional hypothesis for power series is satisfied in the analytic special case when the chosen functions are analytically independent and the remaining ones algebraically dependent on them, but it was deliberately stated formally.

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36. Ideal Differentiation and the Different

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E. Noether, Göttingen: Ideal differentiation and the different.

It is shown how the different of an algebraic number field can be regarded as the differential quotient of a defining ideal. That this definition agrees with the usual one follows from structural theorems which are interesting in themselves and which, in an analogous sense, constitute a generalization of the Lagrange interpolation formula.

A detailed account is to appear in the *Mathematische Annalen.*

37. Normal Basis in Fields without Higher Ramification

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By a normal basis of a Galois number field K/k one means a basis of the principal order $\mathfrak{O}$ of K, relative to the principal order $\mathfrak{o}$ of k, consisting of conjugate elements. A necessary condition for the existence of such a basis is, as is known, that no higher ramification fields occur in K/k. This remains true even if one restricts to a place, i.e. passes to the p-adic extension k_p of k and the corresponding K_p of K. I prove below the converse:

For every place $\mathfrak{p}$ not dividing the degree n of K/k, a normal basis exists. In particular: if K/k has no higher ramification, then at every ramified place a normal basis exists; the discriminant there is the square of a group determinant.

To this last assertion one must add that, contrary to a conjecture of E. Artin, the passage to his conductors still requires further considerations: decomposing the group determinant according to irreducible characters yields generalized root numbers; a suitable grouping yields a factorization in the ground field enlarged by the characters. In the simplest cases I can identify these new factors with Artin's conductors by means of the ideal-theoretic decomposition of the root numbers. In the general case, where no normal basis exists, a splitting into generalized root numbers is still always possible by means of a composition series by Galois modules; from this one again obtains a corresponding decomposition in the enlarged ground field.

The proof of the existence of the normal basis rests on the fact that $\mathfrak{O}/\mathfrak{o}$, viewed as a Galois module, becomes operator-isomorphic to a one-sided ideal of the integral group ring extended by $\mathfrak{o}$. At all ordinary ramification places this group ring is a maximal order; ideals in maximal orders of p-adic semisimple systems are principal ideals. Returning from this to the Galois module gives precisely the existence of the normal basis. At the higher ramification places, the integral group ring passes to a nonmaximal order, and the ideal corresponding to $\mathfrak{O}/\mathfrak{o}$ becomes a nonprincipal ideal.

All these considerations remain valid for function fields of one variable, provided that the characteristic of the constant field does not divide the degree of K/k.

§1. The p-adically Extended Integral Group Ring

Let $\mathfrak{O}$ and $\mathfrak{o}_p$ denote the principal orders of an algebraic number field k and of its p-adic extension k_p, where p is a prime ideal of k. Let (G) denote the rational integral group ring and $[G]$ the integral group ring of a group G with n elements. By $(G)_k$, respectively $(G)_{k_p}$, we mean the group ring with coefficients in k, respectively k_p; analogously $[G]_\mathfrak{o}$ and $[G]_{\mathfrak{o}_p}$ are the corresponding extensions of the integral group ring to coefficients in $\mathfrak{o}$, respectively $\mathfrak{o}_p$.

Theorem 1. If p does not divide the number n of elements of G, then $[G]_{\mathfrak{o}_p}$ is a maximal order of $(G)_{k_p}$. If p divides n, then $[G]_{\mathfrak{o}_p}$ is not maximal.

The discriminant of the integral group ring, and hence of $[G]_\mathfrak{o}$ and $[G]_{\mathfrak{o}_p}$, is a power of n. Thus if p does not divide n, the discriminant of $[G]_{\mathfrak{o}_p}$ is a unit. This means that, while $\mathfrak{o}_p$ is fixed, $[G]_{\mathfrak{o}_p}$ cannot be enlarged to a larger order; such an enlargement would split off a nonunit square factor from the discriminant. This proves the first part.

If, on the other hand, p divides n, the direct sum corresponding to the identity representation,

$$E^0[G]_{\mathfrak{o}_p} + (1 - E^0)[G]_{\mathfrak{o}_p}, \qquad E^0 = \frac{1}{n}\sum_{S\in G} S,$$

is a proper extension of $[G]_{\mathfrak{o}_p}$.

Theorem 2. If p does not divide n, every integral or fractional ideal with respect to $[G]_{\mathfrak{o}_p}$ is principal. Indeed, by Theorem 1, $[G]_{\mathfrak{o}_p}$ is a maximal order in a p-adic semisimple system, and the ideals of such orders are principal.

§2. Galois Modules, Operator Isomorphism, Normal Basis

Let K/k be Galois with group G. For $z \in K$, let z^S denote the element obtained from z by the substitution $S \in G$. The substitutions in G induce operator automorphisms of K/k, regarded as a k-module. Thus K/k can be made into a module with respect to $(G)_k$ by setting

$$z\Big(\sum_i c_i S_i\Big) = \sum_i c_i z^{S_i}, \qquad z \in K,\ c_i \in k,\ S_i \in G.$$

Definition. Every $(G)_k$-module contained in K/k is called a rational Galois module. Similarly, the $[G]_{\mathfrak{o}}$-modules contained in K/k are called integral Galois modules; in particular $\mathfrak{O}/\mathfrak{o}$ is an integral Galois module.

Theorem 3. As a rational Galois module, K/k is operator-isomorphic to $(G)_k$. This follows from the fact that K/k always has a field normal basis, i.e. a k-basis consisting of conjugates z^S. The correspondence

$$\sum_i c_i S_i \longmapsto \sum_i c_i z^{S_i}$$

is an operator homomorphism, and since the ranks are equal it is an isomorphism.

Theorem 4. As an integral Galois module, $\mathfrak{O}/\mathfrak{o}$ is operator-isomorphic to a $[G]_{\mathfrak{o}}$-module in $(G)_k$ of maximum rank n. Likewise, $\mathfrak{O}_p/\mathfrak{o}_p$ is operator-isomorphic to a $[G]_{\mathfrak{o}_p}$-module of rank n in $(G)_{k_p}$.

The isomorphism of Theorem 3 assigns to the $[G]_{\mathfrak{o}}$-module $\mathfrak{O}/\mathfrak{o}$ a $[G]_{\mathfrak{o}}$-module of rank n in $(G)_k$. Locally one extends coefficients from k to k_p; the resulting system is semisimple, in general a sum of isomorphic fields.

Theorem 5. At every place p which does not divide the degree n of K/k, the module $\mathfrak{O}_p/\mathfrak{o}_p$ has a normal basis.

By Theorem 1, $[G]_{\mathfrak{o}_p}$ is a maximal order there; the $[G]_{\mathfrak{o}_p}$-modules of rank n in $(G)_{k_p}$ are integral or fractional ideals, hence principal by Theorem 2. If the ideal corresponding to $\mathfrak{O}_p/\mathfrak{o}_p$ is $W[G]_{\mathfrak{o}_p}$, then it has the $\mathfrak{o}_p$-basis WS_i. The operator isomorphism says that w^{S_i} is an $\mathfrak{o}_p$-basis of $\mathfrak{O}_p/\mathfrak{o}_p$, where w is the element corresponding to W. Thus the conjugate elements w^{S_i} form a normal basis.

If p divides n, the corresponding module cannot be principal, since in this case no normal basis can exist.

Additional remark to Theorem 3. From the operator isomorphism between $(G)_k$ and K/k follow Speiser's results on Klein's problem of forms: every representation of the Galois group G of K/k which is irreducible over k is generated by an irreducible rational Galois module from K/k; every absolutely irreducible representation is generated by a Galois module from K_Z/Z, where Z is a splitting field of the corresponding component of the group ring. If $v_1, \ldots, v_m$ is a basis of such a Galois module and $S \mapsto (s_{ij})$ is the corresponding representation, then

$$v_i^S = \sum_j s_{ij} v_j.$$

This is precisely Klein and Speiser's formulation.

§3. The Discriminant as Group Determinant in Fields without Higher Ramification

For decomposing the discriminant it is enough to consider its decomposition at the individual ramified places. In fields without higher ramification, only places with normal basis are involved.

Theorem 6. For Galois fields K/k without higher ramification, the discriminant at every ramified place is the square of the group determinant of the Galois group after replacing the indeterminates by a normal basis. Corresponding to the irreducible representations, the group determinant decomposes into generalized root numbers; the discriminant decomposes into ideals of the ground field enlarged by the characters, with conjugate complex characters corresponding to the same ideals.

With w^S as a normal basis, the discriminant Δ is the square of

$$D = \det(w^{ST})_{S,T\in G}.$$

This is the group determinant with the w^S in place of the indeterminates x_S. If

$$D = D_1 \cdots D_h$$

is the factorization according to the absolutely irreducible representations, then for a representation $S \mapsto \rho_i(S)$

$$D_i = \det\Big(\sum_{S\in G} w^S \rho_i(S)\Big).$$

Thus the D_i are generalized root numbers. For cyclic groups they are simply the Lagrange root numbers

$$\sum_{S\in G} w^S \chi_i(S).$$

To pass from the factorization of the group determinant to that of the discriminant, one groups each representation with its adjoint. If D_i and D_i' are the corresponding factors, put

$$\Delta_i = D_i D_i'.$$

Then $\Delta_i^T = \Delta_i$ for all $T \in G$; hence Δ_i lies in the ground field enlarged by the real subfield of the i-th character. The discriminant factorization

$$\Delta = \Delta_1 \cdots \Delta_h$$

is therefore a factorization in the field obtained by adjoining the real subfields of the characters, with conjugate complex characters corresponding to the same factors.

If one can show that the ideals generated by the Δ_i actually already lie in the base field k, and are equal for conjugate characters, then they agree with Artin's conductors, provided that to composite characters one assigns the corresponding products of powers of the Δ_i.

Theorem 7. Let k be the field of rational numbers, and let K/k be cyclic of prime degree l, prime to the discriminant, hence without higher ramification. Then the ideals generated by the Δ_λ for the nontrivial representations are all equal and agree with the conductor of K/k.

This follows from the known decomposition of root numbers. At a ramified place p of K/k, for k_ε, the field of l-th roots of unity, and K_ε, one has

$$(p) = \mathfrak{p}_1 \cdots \mathfrak{p}_{l-1}, \qquad \mathfrak{p}_i = \mathfrak{P}_i^l,$$

with $\mathfrak{p}_i$ in k_ε and $\mathfrak{P}_i$ in K_ε. Further, for the root numbers,

$$(\Omega_\lambda) = \mathfrak{P}_1^{r_1} \cdots \mathfrak{P}_{l-1}^{r_{l-1}}, \qquad (\Omega_{\bar\lambda}) = \mathfrak{P}_1^{l-r_1} \cdots \mathfrak{P}_{l-1}^{l-r_{l-1}},$$

where $r_1, \ldots, r_{l-1}$ is a permutation of $1, \ldots, l-1$. The root numbers Ω_λ are here identical with the factors D_λ of the group determinant; hence

$$(\Delta_\lambda) = (D_\lambda D_{\bar\lambda}) = (\Omega_\lambda \Omega_{\bar\lambda}) = \mathfrak{P}_1^l \cdots \mathfrak{P}_{l-1}^l = \mathfrak{p}_1 \cdots \mathfrak{p}_{l-1} = (p),$$

and therefore there is agreement with the conductor of K/k.

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